

THE EXACT BOUND-STATE ANSAETZE
AS ZERO-ORDER APPROXIMATIONS IN PERTURBATION THEORYII. AN ILLUSTRATION: $V(r) = r^2 + fr^2/(1 + gr^2)$

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Challenging Gallas double-well interactions are shown tractable by the quasi Rayleigh Schroedinger perturbation theory of paper I, and a good precision is obtained in the first-order numerical test. The related technicalities are clarified: The existence of the suitable zero-order solutions with $f \neq 0$ and $g \neq 0$ is proved and a few closed formulas for corrections are given and discussed.

1. Introduction

The interaction

$$V(r) = r^2 + fr^2/(1 + gr^2) \quad (1.1)$$

with a strong phenomenological appeal [1] is also a computational challenge [2–8], due to its non-polynomial character. In such a context, our preceding papers [9] and [10] considered a generalization of (1.1) within the framework of perturbation theory. In an unperturbed Schroedinger equation, we replaced the arbitrary (say, phenomenological or “realistic”) shape of $V(r)$ by its suitable Padé approximant.

Both our proposals were based on a common older idea [11] of modification of the textbook Rayleigh-Schroedinger (RS) formalism [12]. In the spirit of Lanczos [13], we introduced a “quasi-unperturbed” basis $|n\rangle$, $n = 0, 1, \dots$ such that a tri-diagonal (or, in general, band-matrix) form of the zero-order Hamiltonian H_0 was obtained. Technically, such a weakening of global information about H_0 did not cause any problems: In the formulas, the “unperturbed” spectrum of H_0 was replaced by certain auxiliary continued fractions (see below).

The key difference between these alternative perturbation prescriptions (namely between the generalized RS, GRS formalism of Ref. [9], and its more conventional, quasi-RS descendant of Ref. [10]) lies in the zero-order approximations. In the former case, a full freedom of choosing the energy E_0 leads to a “democratic” character of separate corrections: Even the zero-order GRS state $|\psi_0\rangle$ must be expanded into an infinite number of the basis vectors $|n\rangle$.

In an application of the GRS technique to Padé potentials [9], we have entirely overlooked the possible existence [14] of some exceptional, elementary or finite-dimensional bound states $|\psi_0\rangle$. Of course, the existence of these states, i.e., a partial solvability of the pertinent zero-order equations, is still an assumption which is weaker than its RS complete-solvability counterpart [12]. Thus, in paper I of the present series [10], we described a procedure which lies somewhere in between the two (namely RS and GRS) extremes. In this quasi-RS formalism, just a single eigenstate of H_0 is needed.

A priori, the quasi-RS approach profits from the drastic simplification of the GRS formulas. At the same time, the quasi-RS technique might improve the classical RS perturbation convergence, due to the extension of class of its "admissible" H_0 's. In such a setting, a numerical study of the quasi-RS formalism seems relevant, and the paper of Gallas [8] also points at a suitable class of examples.

In Section 2 we notice that Gallas potentials (Eq. (1.1) with a special choice of couplings) possess a double-well structure. Simultaneously, they remain tractable as certain generalized harmonic oscillators — within the methodical framework of paper I; this is to be shown in what follows.

Section 3 starts from the zero-order considerations and, in particular, outlines a proof of the existence of $|\psi_0\rangle$ in the required simple form (see also [15]). Then, Section 4 develops the quasi-RS perturbation analysis in more detail. In several versions, the split of the Hamiltonian is discussed, and some explicit formulas are derived. A sample calculation is tabulated and a few complementary remarks are added. The unique role played by the exceptional, isolated zero-order exact solutions is emphasized, and our example (as well as, presumably, all its various Fivel [16] and Coulombic [17] or Padé descendants) is shown to exhibit a "perturbation-friendly" behaviour. Finally, our results are summarized in a very short Section 5.

2. The simplest Padé anharmonicities

A. The quasi-exactly solvable examples

The radial Schroedinger equation

$$\left(-\frac{d^2}{dr^2} + \frac{l(l+1)}{r^2}\right)\psi(r) = E\psi(r) \quad (2.1)$$

$$l = 0, 1, \dots$$

need not necessarily be understood as a definition of F and/or $\psi(E)$. Indeed, in the physics of solitons [18] an important consequence of Eq. (2.1) has been found in the possible "inverse scattering" re-construction of functions $V(r)$ from ψ 's and E 's [19]. An even less sophisticated trick is also used, e.g., in the physics of lasers [20]: We rewrite Eq. (2.1) in the form

$$V(r) = \psi''(r)/\psi(r) - l(l+1)/r^2 + E \quad (2.2)$$

and treat it as a definition of $V(r)$. Of course, the value of $V(0)$ (i.e. the origin of the energy scale) may be chosen arbitrarily.

In accord with Lemma 3 of Ref. [14], the elementary choice of the wave function

$$\psi(r) = r^{l+1} G(r) \cdot \exp(-r^2/2) \quad (2.3)$$

with a nodeless (i.e., ground-state) polynomial

$$G(r) = 1 + g_1 r^2 + g_2 r^4 + \dots + g_p r^{2p} > 0 \quad (2.4)$$

always generates the (l -dependent) potential

$$V(r) = r^2 + \sum_{m=0}^p f_m r^{2m} / G(r). \quad (2.5)$$

Vice versa, an insertion of this potential in (2.1) gives the partially solvable Schroedinger equation. The particular $p = 2$ example

$$\begin{aligned} V(r) &= r^2 + (f_1 r^2 + f_2 r^4) / (1 + g_1 r^2 + g_2 r^4), \\ f_1 &= (8l + 20) g_2 - 4g_1 - (4l + 6) g_1^2, \\ f_2 &= -2g_2 [4 + (2l + 3) g_1], \\ E &= (1 - 2g_1) (2l + 3) g_1 > -2g_2^{1/2} \end{aligned} \quad (2.6)$$

illustrates the restrictions imposed by Eq. (2.4) on the redundant degrees of freedom in (2.5).

B. The double-well $p = 1$ forces

Our a priori knowledge of the exact bound states (2.3) may be of importance in computations [3, 4]. This has attracted much attention in the literature (see, e.g., Refs. [5] for a review). Here, we shall put $p = 1$ and $g_1 = g > 0$ and $g_2 = 0$ in (2.6) and get the interactions of the type (1.1),

$$\begin{aligned} V(r) &= r^2 + f r^2 / (1 + g r^2), \\ f &= f(g) = -4g - (4l + 6) g^2, \\ E &= (1 - 2g) (2l + 3) \end{aligned} \quad (2.7)$$

for the choice of $V(0) = 0$ (see Fig. 1).

The force (2.7) represents a challenge in perturbation theory. Besides a local extreme at $r = 0$, it also acquires a pair of absolute minima at the nonzero values of the coordinate

$$r^2 = r_0^2 = \{[-f(g)]^{1/2} - 1\} / g \quad (2.8)$$

provided only that the right-hand side expression remains real and positive. Thus, a double-well structure is achieved for most values of the couplings,

$$\begin{aligned} g &> (6^{1/2} - 2)/2 \approx 0.22, \quad l = -1 \\ g &> (10^{1/2} - 2)/6 \approx 0.19, \quad l = 0 \\ g &> (14^{1/2} - 2)/10 \approx 0.17, \quad l = 1 \end{aligned} \quad (2.9)$$

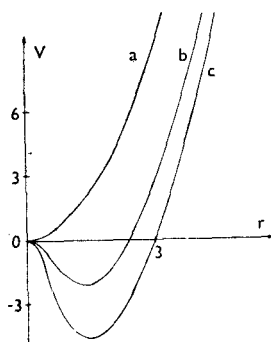


Fig. 1. The shape of the s-wave potential (2.7) with $g = 0$ (a), $g = 0.6$ (b) and $g = 1$ (c).

Let us now shift the potential in such a way that its “capped” form remains non-negative, i.e., $\hat{V}(r) = V(r) - V(r_0) \geq 0$ or $\hat{V}(r_0) = 0$. In the domain of large g 's, this would simplify Figure 1 and gives a very transparent approximate formula

$$\begin{aligned} \hat{V}(r) &= L(L+1)(1/r^2 - 1/r_0^2) + r^2 - r_0^2 + O(1/g), \\ L &= l + 2 + (l + 5/2)m(g), \\ m(g) &= \{1 + 4/[(l + 5/2)^2 g]\}^{1/2} - 1 = O(1/g) \end{aligned} \quad (2.10)$$

which holds for all $r > O(1/g)$. Obviously, we obtain a modified ($L \neq l$) and, of course, completely solvable harmonic oscillator in the limit $g \rightarrow \infty$.

For the finite g 's, we may also replace (2.10) by the exact formulas. For $l = 0$, we have

$$\begin{aligned} \hat{V}(r) &= r^2 + (4 + 6g)/(1 + gr^2) + w(g), \\ w(g) &= 1/g - 2(6 + 4/g)^{1/2}, \quad l = 0 \end{aligned} \quad (2.11)$$

etc. In the new, capped notation, it is easy to notice an almost fixed position of the minima,

$$r_0 = \pm 6^{1/4} + O(1/g) \approx \pm 1.565 + O(1/g), \quad (2.12)$$

as well as a weak g -dependence of the binding energies — we get

$$\hat{E} = \hat{V}(0) + E \approx 2.101 + O(1/g) \quad (2.13)$$

with $l = 0$ in (2.7), etc.

C. The doublets of states in one dimension

In a common one-dimensional reinterpretation of our above differential Schroedinger equation, the choice of integers $l = 0$ and $l = -1$ specifies states with an odd and even parity, respectively. Then, the non-negative polynomial $G > 0$ in (2.4) with $l = 0$ in (2.3) only defines the first excited state. The simultaneous existence of the elementary ground state [Eq. (2.3) with $l = -1$, i.e., $f = -4g - 2g^2$ due to (2.7)] would imply that $g = 0$. Thus, we may conclude that for $p = 1$ (and, mutatis mutandis, for all p 's) a simultaneous existence of the lowest doublet of states would only return us back to the usual $g = 0$ harmonic oscillator.

Recently, a puzzle of the above triviality has been clarified by Gallas [8]. He used the $l = -1$ Ansatz of Blecher and Leach [7]

$$\psi_0(r) = \exp(-r^2/2) (1 + gr^2) \sum_{n=0}^N t_n r^{2n}, \quad (2.14)$$

$$g = g(N)$$

and generalized their explicit algebraic $N \leq 3$ and $l = -1$ constructions of the $g(N)$'s and t_n 's up to $N = 5$. Then, he noticed a regularity in the numerical N -dependence of $g(N)$'s (cf. Table 1 here) and conjectured the existence of "an infinite number of solution pairs" (2.3) and (2.14) with $l = 0$ and $l = -1$, respectively. Here, these solutions have to be employed as the zero-order quasi-RS approximants.

Table 1. Numerical N -dependence of the Gallas couplings $g(N)$.

N	$g(N)$	$g(N) - g(N-1)$
2	1.457 427 108	0.790 8
5	3.783 827 524	0.774 0
10	7.650 218 130	0.773 1
20	15.379 443 684	0.772 9
50	38.564 416 703	0.772 819
100	77.205 163 065	0.772 812 78
200	154.486 331 234	0.772 811 15
500	386.329 577 414	0.772 810 69
1 000	772.734 901 8	0.772 810 63

In accord with Table 2, the restriction $g = g(N)$ in (2.14) makes the minima of our class of potentials practically independent of N . Hence, the d-wave ($L = 2$) harmonic-oscillator approximation (2.10) will hold with a very good precision, and its repulsive core becomes practically fully localized in between the minima (2.12). Its height $\hat{V}(0)$ remains more or less proportional to N (see Table 2 once more).

The strict negativity of the errors makes the related exact $g \gg 1$ oscillator energies

$$4N + 7 + 5\{[1 + 16/(25g)]^{1/2} - 1\} - 2(6 + 4/g)^{1/2} + 1/g \quad (2.15)$$

useful as upper bounds. Indeed, at the large N 's, these bounds apply to pairs of the quasi-degenerate energies (viz., of the $(2N - 1)$ -th and $2N$ -th bound state in the one-dimensional spectrum). Their precision may be estimated, e.g., by comparison with the known energy in (2.14) [8].

Table 2. Minima r_0 and heights of the core $c(N) = \hat{V}(0)$ in the Gallas potentials (2.7).

N	$r_0/6^{1/4}$	$c(N)$	$c(N) - c(N - 1)$
1	0.895 5	2.572	—
2	0.962 9	7.516	4.945
3	0.978 1	12.275	4.758
10	0.994 6	44.924	4.647
100	0.999 51	462.324	4.637

3. The zero-order quasi-RS problem

In terms of the current $g = 0$ harmonic oscillator states $|n\rangle$ with $l = -1$, the Gallas Ansatz (2.14) may be given an alternative form

$$|\psi_0\rangle = (1 + gr^2) \sum_{n=0}^N |n\rangle c_n,$$

$$E_0 = 1 - 6g + 4N, \quad g = g(N). \quad (3.1)$$

This is a typical example of an unperturbed state in the quasi-RS formalism of our preceding paper I. Unfortunately, some of its properties (and, first of all, even its existence) stem from the Gallas conjecture only. Hence, our further considerations must start from a deeper analysis of these technicalities.

A. A numerical evaluation of the Gallas couplings

An insertion of Ansatz (3.1) in our differential Schroedinger equation with potential (2.7) leads to the series of requirements

$$\begin{aligned} A_n c_{n-1} + [B_n + (N - n)/g] c_n + C_n c_{n+1} &= 0, \\ A_n &= (N + 1 - n) [n(n - 1/2)]^{1/2}, \quad B_n = (2n + 1/2)(N - n) + 2n - 1, \\ C_n &= (N + 1 - n) [(n + 1)(n + 1/2)]^{1/2} \\ n &= 0, 1, \dots, N \end{aligned} \quad (3.2)$$

which are just $\langle m |$ – projections of the original equation. All the coefficients $A-C$ are g -independent and, for $N > 1$, positive. Thus, we have to solve an $N + 1$ – dimensional generalized eigenvalue problem

$$Qc = yJc \quad (3.3)$$

with the eigenvalues $y = -1/g$ and with the positive and positively semi-definite matrices Q and J , respectively.

In the first step, we may notice that $J_{NN} = 0$ and reinterpret the g -independent last row in (3.3),

$$A_N c_{N-1} + B_N c_N = 0 \quad (3.4)$$

as a definition of c_N

$$c_N = -c_{N-1} [N(N-1/2)]^{1/2} / (2N-1) \neq 0. \quad (3.5)$$

After we eliminate this value from Eq. (3.2), we get its N -dimensional equivalent

$$\begin{bmatrix} (B_0 - w_0) & C_0 & & \\ A_1 & (B_1 - w_1) & C_1 & \\ & \dots & \dots & \\ & A_{N-2} & (B_{N-2} - w_{N-2}) & C_{N-2} \\ & & A_{N-1} & (G_{N-1} - w_{N-1}) \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \dots \\ c_{N-2} \\ c_{N-1} \end{bmatrix} = 0. \quad (3.6)$$

The new equation contains the non-zero diagonal g -dependent components $w_k = -(N-k)/g$, and the “effective” matrix element

$$G_{N-1} = B_{N-1} - C_{N-1}(1/B_N) A_N = 3N - 9/2 \quad (3.7)$$

replaces the original $B_{N-1} = 4N - 9/2$.

In the second step, our new equation (3.6) (i.e., Eq. (3.3) with Q replaced by Q^{eff} and with the semi-definite J replaced by its definite submatrix) may be symmetrized. Indeed, after a redefinition of coefficients in (3.2)

$$d_k = (N-k) c_k \quad (3.8)$$

our equation becomes an ordinary Hermitean eigenvalue problem

$$\begin{bmatrix} (D_0 - y) & F_0 & & \\ F_0 & (D_1 - y) & F_1 & \\ & \dots & \dots & \\ & F_{N-3} & (D_{N-2} - y) & F_{N-2} \\ & & F_{N-2} & (G_{N-1} - y) \end{bmatrix} \begin{bmatrix} d_0 \\ d_1 \\ \dots \\ d_{N-2} \\ d_{N-1} \end{bmatrix} = 0 \quad (3.9)$$

with N real eigenvalues $y = -1/g$. This is an important result. The proof of existence of the Gallas unperturbed states becomes reduced to a proof of negativity of one of the y 's.

In a purely numerical context, the values of $g(N)$ (Table 1) do not contradict the Gallas hypothesis of an asymptotically linear behaviour of $g(N)$. This is also compatible with the leading-order reformulation

$$[\langle m | r^2 | n \rangle + O(N^2)] \begin{pmatrix} c_0 \\ c_1 \\ \dots \end{pmatrix} = [-1/g(N) + 3/2N] \begin{pmatrix} c_0 \\ c_1 \\ \dots \end{pmatrix}$$

of our Eq. (3.3). Here, the smallest eigenvalue of the truncated left-hand side square of r may be expected small, and the resulting estimate $g(N) > (2/3) N$ is also confirmed by Table 1.

B. The proof of existence and uniqueness of the positive $g(N)$ for all $N > 0$

The new matrix elements in (3.9) read

$$\begin{aligned} D_n &= [(2n + 1/2)(N - n) + 2n - 1]/(N - n), \quad G_{N-1} = 3N - 9/2, \\ F_n &= \{[(N + 1 - n)/(N - 1 - n)] [(n + 1)(n + 1/2)]\}^{1/2}, \\ n &= 0, 1, \dots, N - 1 \end{aligned} \quad (3.10)$$

and remain positive for all the dimensions $N > 2$. Unfortunately, this only implies an existence of at least one negative root $g(N) = 1/y$. The proof of the existence of a positive one (needed in the regular potential (2.7)) is nontrivial. Recently, its feasibility has been announced by several authors [15].

The statement is trivial for $N = 1$ and $N = 2$ [7]. We have $g(1) = 2/3$ and the value of $g(2) = 1/2 + (11/12)^{1/2}$ follows from the two-dimensional secular equation

$$\det \begin{pmatrix} 1/g & (3/2)^{1/2} \\ (3/2)^{1/2} & 3/2 + 1/g \end{pmatrix} = 0. \quad (3.11)$$

For the integer $N > 1$, we may treat our $N + 1$ dimensional Gallas-Schroedinger equation (3.2) as a special case of the more general linear problem

$$\begin{aligned} \langle n | [(H_{\text{HO}} - e_{N+1})(1 + gr^2) + 4 + 6gs] \sum_{m=0}^N |m\rangle c_m &= 0 \\ e_{N+1} &= 4N + 5 \end{aligned} \quad (3.12)$$

with the variable $s = L(L + 1)/6$ and with the secular $N + 1$ - dimensional determinant

$$F(s, y) = \det [H_{\text{HO}} - e_{N+1}] (-y + r^2) - 4y + 6s]. \quad (3.13)$$

We have already noticed that the $g \rightarrow \infty$ limit of (3.12) is exactly solvable,

$$[(H_{\text{HO}} - e_{N+1}) + L(L + 1)/r^2] \sum_{m=0}^N r^2 |m\rangle c_m = 0 \quad (3.14)$$

Its three-dimensional harmonic oscillator eigenfunctions and energies

$$|M, L\rangle \approx \sum_{m=0}^N r^2 |m\rangle c_m, \\ e_{N+1} = 4N + 5 = 4M + 2L + 3 \quad M = 0, 1, \dots \quad (3.15)$$

with an odd integer $L = 2j + 1$ must also have a correct threshold behaviour, $\langle r | M, L \rangle \approx r^{L+1}$. This implies that $L > -1$ and we arrive at the $N + 1$ normalizable solutions with

$$M = N - j, \quad L = 2j + 1, \quad j = 0, 1, \dots, N. \quad (3.16)$$

In the $|y| \gg 1$ asymptotic region, we may derive a compact leading-order formula for the secular determinants

$$F(s, y) = [-y + 3(1 - s)/2]^N, \quad |s| \gg 1, \quad |y| \gg 1. \quad (3.17)$$

This leads to the asymptotic representation of the zeros $s(y)$ or $y(s)$ of $F(s, y)$

$$\frac{2}{3}y_j(s) = -s/j^{\text{eff}}(s), \quad (3.18) \\ j^{\text{eff}}(s) \approx j, \quad j = 1, 2, \dots, N, \quad s \gg 1 \\ f^{\text{eff}}(s) \approx N - j, \quad j = 0, 1, \dots, N - 1, \quad s \gg -1.$$

The range of validity of such an estimate may be specified numerically (see Table 3).

Table 3. Numerical values of $j^{\text{eff}}(s)$ in Eq. (3.18).

N	s	j			
		0	1	2	3
2	-20	1.992	0.997		
	-60	1.999	1.000		
3	-20	2.975	1.968	1.000	
	-60	2.995	1.995	1.000	
6	-20	5.889	4.673	3.704	2.845
	-60	5.967	4.907	3.926	2.972
2	20		0.997	1.974	
	60		1.000	1.998	
3	20		1.018	2.034	2.590
	60		1.001	1.993	2.989
6	20		1.224	3.012	4.613
	60		1.013	2.048	3.078

We are prepared to complete the proof. Indeed, the function $F(s, y)$ is a determinant of a sum of matrices. As a consequence of their (semi-) positivity, its zeros $s(y)$ and/or

$y(s)$ must be non-increasing functions of y and/or s , respectively [21]. Hence, the uniqueness of $g(N) > 0$ follows from our exact knowledge of the complete spectrum at $y = 0$,

$$s = s_j(0) = (j + 1)(2j + 1)/3, \quad j = 0, 1, \dots, N. \quad (3.19)$$

A typical y -dependence of $s(y)$'s is shown here in Fig. 2.

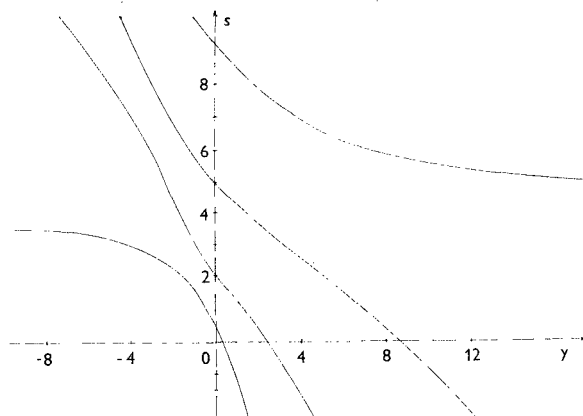


Fig. 2. The zeros of the secular determinant (3.13) for $N = 3$.

4. The quasi-RS perturbation series

A. Perturbations

Interaction (1.1) inspired several RS perturbative studies [2] of Schroedinger equations. In the double-well regime, we may expect a worsening of the RS convergence, and the quasi-RS technique of our preceding paper I may be recommended.

In accord with the detailed quasi-RS prescription, we have to start from a re-defined wavefunction $|\varphi\rangle = G^{-1}|\psi\rangle$. In the simplest case, we may introduce a quantity $h = [f - f(g)]/g$ as a small perturbation parameter. Then, our full Schroedinger equation acquires the almost elementary form

$$(T_0 + hT_1)|\varphi\rangle = E G|\varphi\rangle \quad (4.1)$$

with

$$T_0 = (H_{\text{HO}} - z_0)G + z_0, \quad T_1 = -gr^2. \quad (4.2)$$

We have $z = f/g = z_0 + h$ while H_{HO} denotes the $g = 0$ harmonic oscillator Hamiltonian.

The formal quasi-RS solutions

$$\begin{aligned} E &= E_0 + hE_1 + h^2E_2 + \dots \\ |\varphi\rangle &= |\varphi_0\rangle + h|\varphi_1\rangle + h^2|\varphi_2\rangle + \dots \end{aligned} \quad (4.3)$$

of our Schroedinger Eq. (4.1) may start from the zero-order wave function (2.3) with the odd parity, in full accord with the general formulas of paper I.

The even-parity choice (3.1) of E_0 and $|\varphi_0\rangle$ becomes eligible for $g \approx g(N)$ and $f \approx f(g)$ at some integer $N > 0$. In such a case, we still may vary both g and f in a search for the "best" approximation to a "realistic" interaction $V(r)$. Nevertheless, for the sake of simplicity, we shall keep the odd solution (2.3) unchanged, and fix the coupling $f = f(g)$. Then, the perturbations of the Gallas states (3.1) will be caused by a variation of $g = g(N) + h$ with a small $h \neq 0$.

In the latter case, the unperturbed Schroedinger equation reads

$$(T_0 + hT_1 + h^2T_2)|\varphi\rangle = E(G_0 + hG_1)|\varphi\rangle \quad (4.4)$$

and contains the infinite-dimensional matrices

$$\begin{aligned} G &= G_0 + hG_1, \quad G_0 = 1 + g(N)r^2, \quad G_1 = r^2 \\ T &= T_0 + hT_1 + h^2T_2, \quad T_0 = H_{\text{HO}} + (H_{\text{HO}} - z_0)g_0r^2, \\ z_0 &= 4 + 6g(N), \quad T_1 = [H_{\text{HO}} - z_0 - 6g(N)]r^2, \quad T_2 = -6r^2. \end{aligned} \quad (4.5)$$

Its less trivial structure makes it more suitable for the present illustrative purposes.

B. The energies and numerical tests

All of the separate quasi-RS corrections are, in accord with I, generated by very similar finite-dimensional equations. In particular, the first order Schroedinger equation (4.4) + (4.5) may be given the form of Eq. (4.6) in I with $k = 1$.

$$\begin{aligned} R_0|\varphi\rangle - E_1G_0|\varphi_0\rangle &= |\tau\rangle \\ |\tau\rangle &= (E_0G_1 - T_1)|\varphi_0\rangle = [e_{N+1} - H_{\text{HO}} + 6g(N)]r^2|\varphi_0\rangle. \end{aligned} \quad (4.6)$$

In accord with Eq. (3.8) of I, the tridiagonal matrix $R_0 (= T_0 - E_0G_0)$ possesses a left eigenvector $\langle\chi|$ such that

$$\langle\chi|R_0 = \kappa\langle N+1|, \quad \kappa = \langle\chi|N\rangle\langle N|R_0|N+1\rangle. \quad (4.7)$$

Moreover, the $\langle m|$ -projections of the vector $|\tau\rangle$ in (4.6) become equal to zero for all the integers $m > N+1$. As a direct consequence of these two properties, the normalization of the wave function $v = \langle N+1|\varphi_1\rangle$ and the value of its energy E_1 may be defined by the simple pair of equations

$$\begin{aligned} \langle\chi|R_0|N+1\rangle v &= \langle\chi|\tau\rangle + \langle\chi|G_0|\varphi_0\rangle E_k, \\ \langle N+1|R_0|N+1\rangle v &= \langle N+1|\tau\rangle + \langle N+1|G_0|\varphi_0\rangle E_k \end{aligned} \quad (4.8)$$

(Eqs. (6.1) and (6.4) in I) with $k = 1$ and with an obvious solution.

In the light of Eq. (5.7) of I, an evaluation of the auxiliary analytic continued fractions [22]

$$\begin{aligned} F_m &= 1/(A_m - B_m F_{m+1} C_{m+1}), \quad m \ll M \\ A_m &= \langle N + m | R_0 | N + m \rangle, \quad B_m = \langle N + m | R_0 | N + m + 1 \rangle, \\ C_{m+1} &= \langle N + m + 1 | R_0 | N + m \rangle, \quad F_{M+1} = 0, \quad M \rightarrow \infty \end{aligned} \quad (4.9)$$

simplifies the wave functions,

$$\langle N + k | \varphi_1 \rangle = v \prod_{m=2}^k (-F_m C_m), \quad k > 1. \quad (4.10)$$

Hence, the explicit generation of all the first-order approximants in the Ansatz (4.3) remains elementary.

For a test and verification of the whole quasi-RS perturbation algorithm, let us now solve the pair of relations (4.8) numerically. A sample of results is presented in Table 4. We have chosen $N = 3$ and see that the whole perturbation method works

Table 4. Typical first-order quasi-RS results.

$h/100$	$E_0 + hE_1$	E_{exact}
-3.0	-0.257	-0.260
-2.0	-0.308	-0.310
-1.0	-0.359	-0.359
-0.3	-0.394	-0.394
-0.2	-0.399	-0.399
-0.1	-0.404	-0.404
+0.1	-0.414	-0.414
+0.2	-0.419	-0.419
+0.3	-0.424	-0.424
+1.0	-0.460	-0.459
+2.0	-0.510	-0.509
+3.0	-0.561	-0.558

well even in its first nontrivial approximation. We may only note that for comparison, the exact energies were computed by the numerical method of Ref. [6]. It guarantees their precision via a simultaneous specification of their upper and lower bounds.

C. A few additional remarks

The implicit specification (4.8) of the second-order ($k = 2$) energies will contain a different auxiliary vector,

$$\begin{aligned} |\tau\rangle &= (E_1 + 6) r^2 |\varphi_0\rangle \\ &+ E_1 |\varphi_1\rangle + [e_{N+1} - H_{\text{HO}} + (E_1 + 6) g(N)] r^2 |\varphi_1\rangle. \end{aligned} \quad (4.11)$$

Its second and third components already contain continued fractions (4.9). Their emergence may be seen as analogous to the presence of the full unperturbed spectrum in the ordinary RS second-order corrections [12].

The same (continued fractions $\Leftrightarrow$ unperturbed spectrum) analogy concerns also the higher-order corrections and/or the general $p > 1$ cases. The corresponding matrix continued-fraction quantities remain defined by the $M \rightarrow \infty$ limiting transition [9, 10]. Obviously, the proof of their convergence is necessary. In our $p = 1$ example, such a proof is very easy and may be found, e.g., in Ref. [6].

In the standard perturbative context, the most relevant problem of convergence concerns the expansions (4.3) themselves. For such a purpose, our present example is not the best test – it already gives a very good lowest-order precision. For the

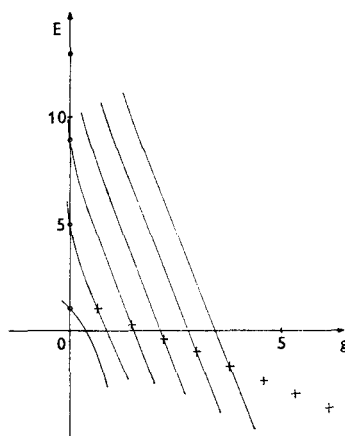


Fig. 3. Exact numerical energies and their harmonic oscillator ($\circ$) and Gallas ($+$) spectra.

explanation of such a phenomenon, let us take the exact numerical energies as functions of the variable g (cf. Fig. 3) and interpolate these quantities by the quasi-perturbative formula

$$E \approx E_{\{m\}0} + hE_{\{m\}1} + \dots + h^m E_{\{m\}m},$$

$$h = g - g(N), \quad m < \infty \quad (4.12)$$

A sample of the resulting coefficients $E_{\{m\}m}$ (Table 5) illustrates the absence or at least the strong suppression of the “higher-order” quasi-perturbation contributions, especially at higher N ’s.

The exceptionality of our present $p = 1$ example lies in a commutativity of some operators in the Schroedinger equations

$$[(H_{\text{HO}} - E - z)(1 + gr^2) + z] \sum_{m=0}^{\infty} |m\rangle \langle m| \varphi\rangle = 0. \quad (4.13)$$

This leads to a trivial possibility of their symmetrization

$$\sum_{m=0}^{\infty} \langle n | [1 + g r^2 + z/(H_{\text{HO}} - E - z)] | m \rangle \langle m | \varphi \rangle = 0 \quad (4.14)$$

for all $n \neq N + 1$ and $|g - g(N)| \ll 1$ at least. Thus, the symmetric matrices

$$\begin{aligned} Q_{jk} &= \langle N + 1 + j | [1 + g(N) r^2 + z_0/(H_{\text{HO}} - e_{N+1})] | N + 1 + k \rangle, \\ j, k &= 1, 2, \dots \\ P_{jk} &= \langle j | [1 + g(N) r^2 + z_0/(H_{\text{HO}} - e_{N+1})] | k \rangle, \\ j, k &= 0, 1, 2, \dots, N \end{aligned} \quad (4.15)$$

may replace the respective non-symmetric submatrices of R_0 [see also our above transition from (3.6) to (3.9)].

Table 5. A sample of coefficients $E_{\{m\}m}$ in (4.12).

N	m	g - g(N)			
		-0.02	-0.01	0.00	0.01
1	0	1.09	1.04	1.00	0.96
	1	-4.38	-4.37	-4.36	
	2		0.58	0.52	0.46
	3		-2.07	-2.02	
2	0	0.35	0.30	0.26	0.21
	1	-4.68	-4.68	-4.69	
	2	-0.19	-0.20	-0.21	
	3		-0.37	-0.36	
4	0	-0.96	-1.01	-1.06	-1.11
	1	-5.16	-5.16	-5.16	
	2		-0.15	-0.15	-0.15
	3		-0.06	-0.06	
9	0	-4.15	-4.21	-4.26	-4.32
	1	-5.57	-5.57	-5.57	
	2		-0.05	-0.05	-0.05
	3		-0.005	-0.005	

Last but not least, we may put $|\chi\rangle = |\varphi_0\rangle$ in all the symmetrized formulas of I for $p = 1$. In our particular example (4.4), such a trick provides the first-order quasi-RS energies in a compact form

$$\begin{aligned} E_1 + 6 &= \langle \varphi_0 | [6(H_{\text{HO}} - e_{N+1})^{-1} + r^2] | \varphi_0 \rangle \\ &\times \{ \langle \varphi_0 | (H_{\text{HO}} - e_{N+1})^{-1} (1 + g(N) r^2) | \varphi_0 \rangle \\ &- [\langle \varphi_0 | N \rangle \langle N | r^2 | N + 1 \rangle g(N)]^2 / z_0 \}^{-1}. \end{aligned} \quad (4.16)$$

This formula is almost as simple as an ordinary RS prescription. The related fixed quasi-RS normalization of the wave functions

$$v = \langle N + 1 | \varphi_1 \rangle = (E_1 + 6) g(N) \langle N + 1 | r^2 | N \rangle \langle N | \varphi_0 \rangle / z_0 \quad (4.17)$$

also remains compact.

5. Summary

(i) The family of our illustrative zero-order “quasi-exact” states (3.1) resembles harmonic oscillators. Indeed, the integers $2N$ specify the number of nodes in $\psi(r)$, for $N \leq 5$ at least [8]. Moreover, the N -dependence of g 's and potentials is so smooth that the Gallas states (crosses in Figure 3) seem to form an equidistant set on a line in the coupling-energy plane. Thus, we may consider them as a set of approximants which deviate very much from the $g = 0$ harmonic oscillators (marked by circles in the same Figure). In the other words, our formalism uses the Gallas states just as certain reliable approximants in a seemingly “strongly non-perturbative” regime.

(ii) In principle, we could relate the Gallas states also to the second, $g \rightarrow \infty$ exactly solvable harmonic oscillator extreme. Also in this degenerate and perturbatively singular domain, the present quasi-RS strategy seems extremely promising, as its “standard” RS treatment remains entirely unclear.

(iii) Most of the Gallas states seem to lie in a domain with an almost linear g - or h -dependence of energies. Table 5 confirms this observation quantitatively. Thus, we may even expect an improvement of the quasi-RS convergence for the higher excitations N . This contrasts with an increasing significance of the higher RS corrections in the vicinity of $g = 0$, and provides an additional, purely numerical motivation for a preference of the quasi-RS formalism.

(iv) In general, the failure of various perturbative computations may at least sometimes be assigned to an inadequate split

$$H = H_0 + hH_1 + \dots \quad (5.1)$$

of the Hamiltonians in question. We are, of course, limited by the scarcity of the available, exactly solvable H_0 's. Thus, for all the Hamiltonians which lie “too far” from these H_0 's, the RS energies and wave functions may diverge even for an arbitrarily small coupling or measure h of the perturbations [23]. Vice versa, the quasi-RS rearrangement of (5.1) with a “better”, incompletely solvable zero-order approximant H_0 may improve the convergence of solutions. This was our basic idea — our examples support its efficiency and recover also a gratifying formal parallelism between the explicit RS and quasi-RS formulas.

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